Classical inequalities

L4-AL2 Level 4 Olympiad · Algebra

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The standard olympiad inequalities: AM-GM, Cauchy-Schwarz and its Engel form, rearrangement, homogenising, SOS and the tangent line trick, with proofs written to BMO standard.
By the end you can choose between AM-GM, Cauchy-Schwarz, rearrangement and Chebyshev, splitting terms so that the equality case is attainable; homogenise a constrained inequality and use SOS or the tangent line trick; write an inequality proof that states every tool, every positivity condition and the equality case.
Prerequisites: Factorising, substitution and systems (L2-AL1), Inequalities and optimisation (L3-AL2). **Contests:** BMO Round 1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Find the minimum of $(a+b)\left(\frac{1}{a}+\frac{4}{b}\right)$ over positive reals a, b.

A first attempt is AM-GM on each bracket: $a+b \geq 2\sqrt{ab}$ and $\frac{1}{a}+\frac{4}{b} \geq 2\sqrt{\frac{4}{ab}} = \frac{4}{\sqrt{ab}}$, so the product is at least 8. Each step is correct, but 8 is not the minimum. Equality in the first step needs $a = b$, and in the second needs $\frac{1}{a} = \frac{4}{b}$, that is $b = 4a$. These cannot both hold, so the bound 8 is never reached.

Expanding instead gives $5 + \frac{4a}{b} + \frac{b}{a} \geq 5 + 2\sqrt{4} = 9$, with equality when $b = 2a$. So the minimum is 9. The principle behind this unit is that *every inequality step carries an equality condition, and a chain of steps is sharp only if all of those conditions can hold at once.* Hence one should find the equality case first, then choose inequalities that are tight there.

1.1 AM-GM

AM-GM.
For non-negative reals $x_1, \ldots, x_n$,

$$\frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n},$$

with equality if and only if all the x_i are equal.
Proof. For $n = 2$ this is $(\sqrt{x} - \sqrt{y})^2 \geq 0$. Suppose it holds for n numbers, and take $n+1$ non-negative numbers with mean A. If they are all equal to A we are done. Otherwise one is below A and one is above; relabel so that $x_n \leq A \leq x_{n+1}$. The n numbers $x_1, \ldots, x_{n-1}$, $x_n + x_{n+1} - A$ are non-negative with mean A, so their product is at most A^n. Also $A(x_n + x_{n+1} - A) - x_n x_{n+1} = (A - x_n)(x_{n+1} - A) \geq 0$. Hence

$$x_1 \cdots x_{n+1} \leq x_1 \cdots x_{n-1} \cdot A\,(x_n + x_{n+1} - A) \leq A \cdot A^n.$$

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For the equality case, suppose the numbers are not all equal, so that $A > 0$ and we can choose $x_n < A < x_{n+1}$. If $x_1 \cdots x_{n-1} > 0$, the first inequality is now strict because $(A - x_n)(x_{n+1} - A) > 0$; if $x_1 \cdots x_{n-1} = 0$, the left side is $0 < A^{n+1}$. Either way equality fails.

The main technique in practice is **splitting**: to make AM-GM tight, break terms into equal pieces so that the pieces are equal at the equality case and their product is constant.

Example 1.
Positive reals x, y, z satisfy $xy^2z^3 = 108$. Find the minimum value of $x + y + z$.

Solution.
AM-GM on x, y, z directly gives a bound in terms of xyz, which we do not know. We need the product to be xy^2z^3, so y should appear twice and z three times. Split y into two halves and z into three thirds, giving six terms:

$$\frac{x+y+z}{6} = \frac{x + \frac{y}{2} + \frac{y}{2} + \frac{z}{3} + \frac{z}{3} + \frac{z}{3}}{6} \geq \sqrt[6]{x\left(\frac{y}{2}\right)^2\left(\frac{z}{3}\right)^3} = \sqrt[6]{\frac{108}{108}} = 1.$$

So $x + y + z \geq 6$. Equality needs $x = \frac{y}{2} = \frac{z}{3}$, and combined with the constraint this gives $x = 1, y = 2, z = 3$, where indeed $1 \cdot 4 \cdot 27 = 108$ and $x + y + z = 6$. The minimum is 6.
We split into halves and thirds because the pieces must be equal at the optimum. In general, guess that the pieces $x, y/2, z/3$ are equal at equality and check that the product then comes out constant.

1.2 Cauchy-Schwarz and its Engel form

Cauchy-Schwarz.
For reals a_i, b_i,

$$(a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2) \geq (a_1b_1 + \cdots + a_nb_n)^2,$$

with equality if and only if the vectors (a_i) and (b_i) are proportional.
Proof. If all $a_i = 0$ there is nothing to prove. Otherwise $\sum(a_it - b_i)^2 \geq 0$ for every real t; this is a quadratic in t with positive leading coefficient that is never negative, so its discriminant $4\left(\sum a_ib_i\right)^2 - 4\sum a_i^2 \sum b_i^2$ is at most 0.
Engel form. For reals x_i and positive b_i,

$$\frac{x_1^2}{b_1} + \cdots + \frac{x_n^2}{b_n} \geq \frac{(x_1 + \cdots + x_n)^2}{b_1 + \cdots + b_n},$$

with equality when $x_1/b_1 = \cdots = x_n/b_n$. Apply Cauchy-Schwarz to the vectors $(x_i/\sqrt{b_i})$ and $(\sqrt{b_i})$.

Note that the opening problem takes one line in Engel form: $(a + b)\left(\frac{1^2}{a} + \frac{2^2}{b}\right) \geq (1 + 2)^2 = 9$.

Example 2.
Prove that for positive reals a, b, c,

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$$\frac{a}{2b+3c}+\frac{b}{2c+3a}+\frac{c}{2a+3b}\geq\frac{3}{5}.$$

Solution.
Equality holds at $a=b=c$, where each fraction is $\frac{1}{5}$. AM-GM on the three fractions would need the product of the denominators to be at most $125abc$, which is false in general, so that approach fails. The Engel form needs squares in the numerators, so multiply each fraction by $\frac{a}{a}$, $\frac{b}{b}$, $\frac{c}{c}$:

$$\sum\frac{a^2}{2ab+3ac}\geq\frac{(a+b+c)^2}{(2ab+3ac)+(2bc+3ab)+(2ca+3bc)}=\frac{(a+b+c)^2}{5(ab+bc+ca)}.$$

Finally $(a+b+c)^2\geq 3(ab+bc+ca)$, because the difference is $a^2+b^2+c^2-ab-bc-ca=\frac{1}{2}\left[(a-b)^2+(b-c)^2+(c-a)^2\right]\geq 0$. So the sum is at least $\frac{3}{5}$, with equality exactly when $a=b=c$.

1.3 Rearrangement

Rearrangement.
If $x_1\leq\cdots\leq x_n$ and $y_1\leq\cdots\leq y_n$, then for every permutation σ,

$$x_1y_{\sigma(1)}+\cdots+x_ny_{\sigma(n)}\leq x_1y_1+\cdots+x_ny_n.$$

Similarly ordered sequences give the largest sum, oppositely ordered ones the smallest.
Proof. If $i<j$ but $\sigma(i)>\sigma(j)$, swapping the two values changes the sum by $(x_j-x_i)(y_{\sigma(i)}-y_{\sigma(j)})\geq 0$. Each swap reduces the number of out-of-order pairs, so after finitely many swaps we reach the identity without ever decreasing the sum.
Chebyshev. Pair x_i with y_{i+r} (indices taken mod n) for each shift $r=0,1,\ldots,n-1$. Each sum is at most $\sum x_iy_i$, and adding all n of them gives $n\sum x_iy_i\geq\left(\sum x_i\right)\left(\sum y_i\right)$ for similarly ordered sequences.

Example 3.
Prove that for positive reals a,b,c,

$$a^ab^bc^c\geq a^bb^cc^a.$$

Solution.
AM-GM does not handle products of powers well, especially when the exponents are the variables themselves. Taking logarithms turns products into sums, and since ln is increasing the claim is equivalent to

$$a\ln a+b\ln b+c\ln c\geq b\ln a+c\ln b+a\ln c.$$

Now look at the two sequences (a,b,c) and $(\ln a,\ln b,\ln c)$. Because ln is increasing, they are sorted in the same order, whatever that order is. The left side pairs each number with its own logarithm, which is the similarly ordered pairing. The right side pairs $\ln a$ with b, $\ln b$ with c and $\ln c$ with a, which is another pairing of the same two sequences. By rearrangement the left side is at least the right side, and we are done.

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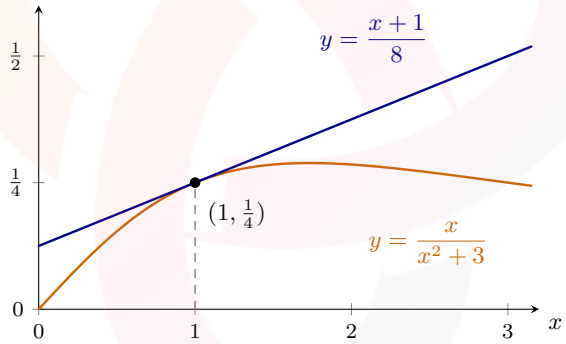
Note that rearrangement needs no hypothesis about which variable is largest, only that the two sequences are sorted the same way, and applying an increasing function to the variables always produces such a pair.

1.4 Homogenising, SOS and the tangent line trick

An inequality is **homogeneous** if every term has the same degree; then you may scale the variables, for instance to make $a+b+c=3$. Conversely, a constraint lets you homogenise: with $a+b+c=3$, the statement $a^2+b^2+c^2 \geq 3$ is equivalent to $a^2+b^2+c^2 \geq \frac{1}{3}(a+b+c)^2$, which is the familiar inequality $3(a^2+b^2+c^2) \geq (a+b+c)^2$.

SOS (sum of squares) means writing the difference of the two sides as $S_c(a-b)^2 + S_a(b-c)^2 + S_b(c-a)^2$ with non-negative coefficients. You have already seen the basic identity in Example 2. The routine for a cyclic sum: subtract the equality value from each term, then regroup by the differences $a-b, b-c, c-a$.

The **tangent line trick** handles $\sum f(a)$ under a constraint like $a+b+c=3$ with equality at $a=b=c=1$. Find a line $g(x) = f(1) + m(x-1)$ with $f(x) \leq g(x)$ (or $\geq$) for every allowed x; then summing gives $\sum f(a) \leq \sum g(a) = 3f(1)$. The slope m is $f'(1)$, or equivalently the value that makes $(x-1)^2$ a factor of the resulting polynomial.



Example 4.
Positive reals a, b, c satisfy $a+b+c=3$. Prove that

$$\frac{a}{a^2+3} + \frac{b}{b^2+3} + \frac{c}{c^2+3} \leq \frac{3}{4}.$$

Solution.
AM-GM gives $a^2+3 \geq 2\sqrt{3}\,a$, so each term is at most $\frac{1}{2\sqrt{3}}$ and the sum is at most $\frac{\sqrt{3}}{2} \approx 0.866$. This is true but too weak, because that AM-GM is tight at $a=\sqrt{3}$, while our equality case is $a=1$.
Instead we use a tangent line at the equality case. Let $f(x) = \frac{x}{x^2+3}$, so $f(1) = \frac{1}{4}$, and look for $f(x) \leq \frac{1}{4} + m(x-1)$. With $m = \frac{1}{8}$ the claim is $\frac{x}{x^2+3} \leq \frac{x+1}{8}$, which (as $x^2+3 > 0$) is equivalent to

$$(x+1)(x^2+3) - 8x = x^3 + x^2 - 5x + 3 = (x-1)^2(x+3) \geq 0,$$

true for all positive x. Summing over a, b, c gives a total of at most $\frac{(a+b+c)+3}{8} = \frac{3}{4}$, with equality at $a=b=c=1$. The picture above shows the line lying above the curve and touching it at $x=1$.

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A second proof, also tight at $a = b = c = 1$: $a^2 + 3 = (a^2 + 1) + 2 \geq 2a + 2$, so each term is at most $\frac{a}{2(a+1)} = \frac{1}{2} - \frac{1}{2(a+1)}$, and Engel form gives $\sum \frac{1}{a+1} \geq \frac{9}{a+b+c+3} = \frac{3}{2}$, so the total is at most $\frac{3}{2} - \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$.

1.5 Writing it up

Markers want to see each named inequality stated with what it is applied to, the positivity that justifies every multiplication or division, and the equality case. For a "find the minimum" problem you need two things: a proof of the bound and an example that attains it. Write the proof forwards. If you discovered it by working backwards, make sure every step reverses, and say so.

A full-marks BMO inequality solution is usually short: one sentence naming the substitution or normalisation (with a reason it is allowed, such as homogeneity), the main estimate with the named inequality, a line deducing the result, and a sentence on when equality holds. Long chains of unexplained algebra lose marks even when they are correct, because the marker cannot see which steps are reversible. If a step uses that a quantity is positive, say why it is positive.

Common traps.

- An equality case that cannot occur, as with the bound 8 in the opening problem, or one that lies outside the allowed range.
- Using AM-GM on quantities that might be negative, or multiplying inequalities whose sides might be negative.
- Working backwards through steps that do not reverse, such as squaring both sides.
- Assuming $a \geq b \geq c$ for an expression that is only cyclic. For cyclic expressions you may assume only that a is the largest.
- A tangent line bound checked near $x = 1$ but not over the whole allowed range.

1.6 How to spot this in a contest

Symmetric expression, a constraint, equality at the centre: think AM-GM, SOS, or a tangent line. Fractions with sums in the denominators: think Engel form, after multiplying up so that the numerators are squares. Products of quantities that are sorted together, such as sides and opposite angles: think rearrangement or Chebyshev. AIME "minimum value" problems are usually AM-GM with a careful split, and the answer is only right if the equality case is attainable.

BMO Round 1 often sets an inequality inside geometry: lengths, areas or ratios in a triangle. Translate into algebra first. For the sides of a triangle, the substitution $a = y + z$, $b = z + x$, $c = x + y$ with $x, y, z > 0$ turns the triangle inequality into plain positivity.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find the minimum value of $x^2 + \dfrac{16}{x}$ over positive reals x.

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Problem 2 ⋆⋆ Positive reals a, b satisfy $a + 2b = 1$. Find the minimum value of $\dfrac{1}{a} + \dfrac{1}{b}$.

Problem 3 ⋆⋆ Find the minimum value of $\dfrac{x^2+3}{\sqrt{x^2+2}}$ over all real x.

Problem 4 ⋆⋆⋆ Positive reals a, b, c satisfy $abc = 1$. Prove that $a^3 + b^3 + c^3 \geq a^2 + b^2 + c^2$.

Problem 5 ⋆⋆⋆ Positive reals a, b, c satisfy $a + b + c = 3$. Prove that

$$\frac{a+3}{a^2+3} + \frac{b+3}{b^2+3} + \frac{c+3}{c^2+3} \leq 3.$$

Problem 6 ⋆⋆⋆⋆ Prove that for all positive reals a, b, c,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2} + \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2(a+b+c)^2}.$$

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME 1983 Problem 9** ⋆⋆
   AM-GM, but check that the equality case is attainable.
2. **BMO Round 1 2002/03 Problem 3** ⋆⋆
   Homogenise using the constraint, then compare with a familiar symmetric inequality.
3. **BMO Round 1 2006/07 Problem 5** ⋆⋆⋆
   Pair the four factors into two products and use AM-GM on the pair.
4. **BMO Round 1 2007/08 Problem 5** ⋆⋆⋆
   Area (barycentric) coordinates turn each ratio into a simple fraction; then AM-GM.
5. **BMO Round 1 2013/14 Problem 6** ⋆⋆⋆
   Rearrangement: in a triangle the longer side faces the larger angle.
6. **BMO Round 1 2010/11 Problem 6** ⋆⋆⋆
   Use the triangle inequality with the constraint to show that each side is less than 1.
7. **USAJMO 2014 Problem 1** ⋆⋆⋆
   The minimum is at most the geometric mean; bound each variable's contribution to the product separately.
8. **USAJMO 2011 Problem 2** ⋆⋆⋆⋆
   Use the constraint to replace the 1 in each numerator by something homogeneous.
9. **USAJMO 2018 Problem 2** ⋆⋆⋆⋆
   Everything is homogeneous: order the variables and use the condition to relate them.
10. **USAJMO 2012 Problem 3** ⋆⋆⋆⋆
    Find a lower bound for each fraction that is a combination of squares (an SOS or tangent line idea).
11. **USAMO 2003 Problem 5** ⋆⋆⋆⋆
    Normalise $a + b + c = 3$, then find a tangent line bound for each term.

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4 Hints

1. Use AM-GM on three terms, splitting $\frac{16}{x}$ into two equal pieces so that the product is constant.
2. Use the Engel form with denominators a and $2b$. What numerator over $2b$ gives back $\frac{1}{b}$?
3. Write the expression in terms of $t = \sqrt{x^2+2}$. Can the obvious AM-GM equality case actually happen?
4. Chebyshev's inequality relates $a^3+b^3+c^3$ to $(a+b+c)(a^2+b^2+c^2)$. Then use the constraint to bound $a+b+c$.
5. Find a linear upper bound for $\frac{x+3}{x^2+3}$ that is tight at $x=1$, and check carefully for which x it holds.
6. Subtract $\frac{1}{2}$ from each fraction and regroup the sum as terms with factors $(a-b)^2$, $(b-c)^2$ and $(c-a)^2$.

5 Solutions

Problem 1

Find the minimum value of $x^2+\dfrac{16}{x}$ *over positive reals* x.

By AM-GM on the three positive terms x^2, $\frac{8}{x}$, $\frac{8}{x}$,

$$x^2+\frac{16}{x} = x^2+\frac{8}{x}+\frac{8}{x} \geq 3\sqrt[3]{x^2\cdot\frac{8}{x}\cdot\frac{8}{x}} = 3\sqrt[3]{64} = 12.$$

Equality needs $x^2=\frac{8}{x}$, that is $x=2$, and indeed $4+8=12$. So the minimum is 12.

Answer: 12

Problem 2

Positive reals a,b *satisfy* $a+2b=1$. *Find the minimum value of* $\dfrac{1}{a}+\dfrac{1}{b}$.

Since $\frac{1}{b}=\frac{(\sqrt{2})^2}{2b}$, the Engel form of Cauchy-Schwarz gives

$$\frac{1}{a}+\frac{1}{b} = \frac{1^2}{a}+\frac{(\sqrt{2})^2}{2b} \geq \frac{(1+\sqrt{2})^2}{a+2b} = 3+2\sqrt{2}.$$

Equality needs $\frac{1}{a}=\frac{\sqrt{2}}{2b}$, that is $2b=\sqrt{2}\,a$. Together with $a+2b=1$ this gives $a=\frac{1}{1+\sqrt{2}}=\sqrt{2}-1$ and $b=1-\frac{\sqrt{2}}{2}$, both positive. Then $\frac{1}{a}=\sqrt{2}+1$ and $\frac{1}{b}=\frac{2}{2-\sqrt{2}}=2+\sqrt{2}$, which sum to $3+2\sqrt{2}$. So the minimum is $3+2\sqrt{2}$.

Answer: $3+2\sqrt{2}$

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Problem 3

Find the minimum value of $\dfrac{x^2+3}{\sqrt{x^2+2}}$ *over all real* x.

Let $t = \sqrt{x^2+2}$, so $t \geq \sqrt{2}$ and the expression is $\frac{t^2+1}{t} = t + \frac{1}{t}$. AM-GM gives $t + \frac{1}{t} \geq 2$, but equality needs $t = 1$, which is impossible since $t \geq \sqrt{2}$. So 2 is not the answer.

Instead compare with the value at $t = \sqrt{2}$:

$$t + \frac{1}{t} - \left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = (t - \sqrt{2}) - \frac{t - \sqrt{2}}{\sqrt{2}\,t} = \frac{(t-\sqrt{2})(\sqrt{2}\,t - 1)}{\sqrt{2}\,t}.$$

For $t \geq \sqrt{2}$ both factors in the numerator are non-negative, so the expression is at least $\sqrt{2} + \frac{1}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$, with equality at $t = \sqrt{2}$, that is $x = 0$. The minimum is $\frac{3\sqrt{2}}{2}$.

Answer: $\frac{3\sqrt{2}}{2}$

Problem 4

Positive reals a, b, c *satisfy* $abc = 1$. *Prove that* $a^3 + b^3 + c^3 \geq a^2 + b^2 + c^2$.

The inequality is symmetric, so we may assume $a \geq b \geq c$. Then also $a^2 \geq b^2 \geq c^2$, so the sequences (a, b, c) and (a^2, b^2, c^2) are similarly ordered. By the rearrangement inequality, pairing them in the same order gives the largest sum, so

$$a^3 + b^3 + c^3 \geq ab^2 + bc^2 + ca^2 \quad \text{and} \quad a^3 + b^3 + c^3 \geq ac^2 + ba^2 + cb^2.$$

Adding these two inequalities to $a^3 + b^3 + c^3 = a^3 + b^3 + c^3$ gives $3(a^3 + b^3 + c^3) \geq (a + b + c)(a^2 + b^2 + c^2)$.

By AM-GM, $a + b + c \geq 3\sqrt[3]{abc} = 3$. Since $a^2 + b^2 + c^2 > 0$, we get $3(a^3 + b^3 + c^3) \geq 3(a^2 + b^2 + c^2)$, which is the required inequality. Equality holds when $a = b = c = 1$.

Problem 5

Positive reals a, b, c *satisfy* $a + b + c = 3$. *Prove that*

$$\frac{a+3}{a^2+3} + \frac{b+3}{b^2+3} + \frac{c+3}{c^2+3} \leq 3.$$

We claim that for $0 < x < 3$,

$$\frac{x+3}{x^2+3} \leq \frac{5-x}{4}.$$

Since $x^2 + 3 > 0$, this is equivalent to $4(x+3) \leq (5-x)(x^2+3)$, that is, to

$$0 \leq -x^3 + 5x^2 - 7x + 3 = (x-1)^2(3-x),$$

which is true because $x < 3$. (The line has slope $-\frac{1}{4}$, which is the derivative of $\frac{x+3}{x^2+3}$ at $x = 1$; this is why $(x-1)^2$ appears.)

Each of a, b, c lies in the interval $(0, 3)$, because the other two are positive and the three add up to 3. Applying the claim to each and adding gives

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$$\sum \frac{a+3}{a^2+3} \le \frac{15-(a+b+c)}{4} = 3.$$

Equality holds when $a=b=c=1$. Note that the linear bound fails for $x>3$, so the range check is an essential part of the proof.

Problem 6

Prove that for all positive reals a, b, c,

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b} \ge \frac{3}{2}+\frac{(a-b)^2+(b-c)^2+(c-a)^2}{2(a+b+c)^2}.$$

First we write the left-hand side minus $\frac{3}{2}$ as a sum of squares. For each fraction,

$$\frac{a}{b+c}-\frac{1}{2}=\frac{2a-b-c}{2(b+c)}=\frac{(a-b)+(a-c)}{2(b+c)}.$$

Adding the three such expressions and collecting the two terms that contain $a-b$, we get

$$\frac{a-b}{2(b+c)}+\frac{b-a}{2(c+a)}=\frac{a-b}{2}\cdot\frac{(c+a)-(b+c)}{(b+c)(c+a)}=\frac{(a-b)^2}{2(b+c)(c+a)}.$$

The terms containing $b-c$ and $c-a$ combine in the same way, so

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}-\frac{3}{2}=\frac{(a-b)^2}{2(b+c)(c+a)}+\frac{(b-c)^2}{2(c+a)(a+b)}+\frac{(c-a)^2}{2(a+b)(b+c)}.$$

Since a, b, c are positive, each of $a+b$, $b+c$ and $c+a$ is less than $a+b+c$, so each product of two of them is less than $(a+b+c)^2$. Hence $\frac{(a-b)^2}{2(b+c)(c+a)} \ge \frac{(a-b)^2}{2(a+b+c)^2}$, with equality only when $a=b$, and similarly for the other two terms. Adding the three bounds gives the required inequality, with equality if and only if $a=b=c$.